



UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 39189

LM PHYSICS OF RENEWABLE ENERGY

SUMMER EXAMINATIONS 2025

Time allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this section.

1. Consider a planet of radius A , albedo α , unit emissivity with no atmosphere. It is illuminated by a star, at a distance R , with a power per unit area (at the planet's distance from the star) I_0 . It has a moon of radius a in a circular orbit around the planet of radius r obeying $A \ll r \ll R$, approximately in the same plane at the planetary orbit. When the moon is at a greater distance than the planet from the star it may be considered to be fully illuminated by the star. The moon has unit albedo and maybe regarded as being at absolute zero temperature. The planet rotates rapidly enough that the night and day temperatures are identical.

How much does the existence of the moon increase the temperature of the planet when the moon is fully illuminated by the star?

[10]

2. The following parts use dimensional analysis.

- (a) The potential energy per wave length, per unit of length of wavefront, $\mathcal{E}$, of water waves has dimensions of MT^{-2} . Find an expression for $\mathcal{E}$ involving the density of water, ρ_w , the amplitude, a , of the waves and the gravitational acceleration, g .

[4]

- (b) Find an expression for the group velocity, v_g , of the water waves assuming it depends on only the wavevector, k , of the waves and the gravitational acceleration, g .

[2]

- (c) Find an expression for the power (per unit length of wavefront), P , assuming it depends only on $\mathcal{E}$ and v_g .

[2]

- (d) Determine the power per unit length of wavefront, P , available from waves assuming a typical amplitude is 1 metre, a typical wavelength is 10 metres and the density of water is approximately 1 kg m^{-3} .

[1]

- (e) The UK extends roughly 1,000 km North to South. How much power might be generated using the above estimate?

[1]

3. Power stations which involve Carnot cycles need plentiful cooling water. Gauging the flow in rivers can be important in this endeavour. One means to perform the measurement is to use a weir, Fig. (1), which is a submerged barrier perpendicular to the direction of flow of the river. Assume the flow is irrotational, steady and the velocity is independent of depth at the point indicated in the figure.

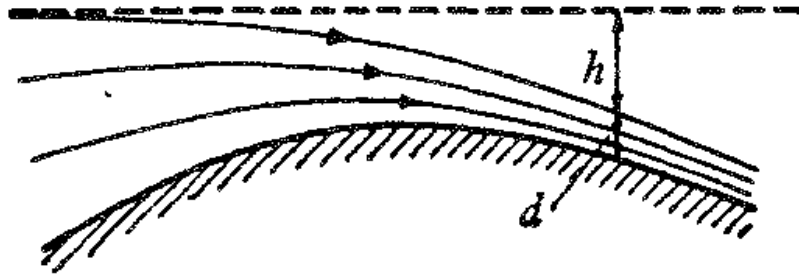


Figure 1: Flow over a weir. At the indicated point, where the depth of the fluid is d , the surface is at a depth h compared to the level far to the left of the figure.

By using Bernoulli's principle determine the speed of flow, v , at the indicated point and hence the rate of flow over the weir per unit length of the weir, Q .

[10]

4. A possible hydropower solution to storing energy for the UK is to hollow out Ben Nevis and fill the resulting cavity with water. Regard Ben Nevis as a cuboid of height $h = 1$ km, and area $A = 10 \text{ km} \times 10 \text{ km}$.

(a) Estimate the potential energy which could be stored, using the density of water as approximately 10^3 kg m^{-3} .

[4]

(b) The UK electricity demand is of the order of 30 GW. If there was a spell of two weeks with no wind or solar power and no natural gas burnt (i.e 85% of supply non-existent), would this hydro project suffice (assuming perfect efficiency of electricity generation)?

[2]

(c) Estimate the cost of construction, noting the historical cost of $\pounds 6 \times 10^9$ to excavate $2 \times 10^5 \text{ m}^3$ for the Dinorwig hydro station which took ten years to construct.

[2]

(d) Estimate the cost of material for constructing (on a flat site) a reservoir with concrete walls containing the same volume of water. A rule of thumb is that a dam has to have a thickness of half its height at its base. The cost of concrete is roughly $\pounds 100 \text{ m}^{-3}$. The UK GDP is roughly $\pounds 3 \times 10^{12}$.

[2]

SECTION 2

Answer **two** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

5. Consider a region of the surface of a planet which may be regarded as flat. Black body radiation from a star is absorbed with a power per unit area $(1 - A)I$, where $0 < A < 1$ is the albedo of the planet and I is the incident flux of radiation. The planet radiates, as a black body with unit emissivity, at a temperature T_0 , which is much lower than that of the star.

There is a gaseous layer above the planet which is completely transparent to the radiation from the star, but completely absorbing the black body radiation from the surface of the planet. It re-radiates the energy as a black body (with unit emissivity) with temperature T_g .

- (a) Show that the temperature at the surface of the planet is:

$$T_0 = \left(\frac{2(1 - A)I_0}{\sigma} \right)^{1/4},$$

where σ is the Stefan-Boltzmann constant.

[7]

It has been proposed that enhancing the albedo of the planet might reduce the surface temperature. Consider modifying the gaseous layer to have a non-zero albedo $0 < a < 1$ for frequencies of the radiation from the star, with $1 - a$ of the radiation transmitted by the layer. The gas remains completely absorbing at the frequencies of the black-body radiation from the planet.

- (b) Show that the **total albedo**, A^t , of the planet and gaseous layer, allowing for multiple reflections, is

$$A^t = a + \frac{(1 - a)^2 A}{1 - aA}.$$

[8]

- (c) Show that $A^t > A$.

[5]

- (d) Calculate the temperature, $\tilde{T}_0$, of the surface of the planet.

[6]

- (e) Show that $\tilde{T}_0 < T_0$.

[4]

6. A wind turbine situated at $x = 0$ experiences an airflow parallel to the x -axis, with velocity $U\hat{\mathbf{x}}$ as $x \rightarrow -\infty$ and fU as $x \rightarrow \infty$, where $0 < f < 1$. The turbine rotates, with angular velocity $\Omega = \Omega\hat{\mathbf{x}}$ in the perpendicular, y - z plane. The pressure as $x \rightarrow \pm\infty$ is p_0 . Air may be assumed to have no viscosity and be incompressible with density ρ_0 . The flow is irrotational for the purposes of this question.

- (a) By appealing to a general principle explain why the velocity at the centre of the turbine, u_T , is the same on both sides of the turbine. [3]

- (b) If the pressure in the centre of the turbine on the $x < 0$ side is p_- and that on the $x > 0$ side is p_+ , show by using Bernoulli's principle that: [7]

$$p_- - p_+ = \frac{1}{2}(1 - f^2)U^2 .$$

- (c) Noting that the difference between the momentum flux in the x -direction, Δj_x^{mom} at $x = \pm\infty$ is $\rho_0 U(1 - f)$, deduce an expression for the power, P , of the wind turbine. Deduce another expression for the power from the previous part of this question. From these two expressions show that [5]

$$u_T = \frac{1}{2}(1 + f)U .$$

The wind turbine has N equally-spaced blades of length ℓ . They all present the same angle of attack, α , along their length to the "effective wind", $\mathbf{u}^{\text{eff}}(\mathbf{r})$, a sum of the real wind at the turbine, $u_T\hat{\mathbf{x}}$ and the effect of rotation:

$$\mathbf{u}^{\text{eff}}(\mathbf{r}) = u_T\hat{\mathbf{x}} - \Omega \times \mathbf{r} ,$$

where $\mathbf{r}$ denotes the centre of mass of the cross section of the blade, with $0 \leq r \leq \ell$, and $r = |\mathbf{r}|$. It is convenient to define the angle between $\mathbf{u}^{\text{eff}}(\mathbf{r})$ and $\Omega \times \mathbf{r}$ to be $\phi(r)$, where $\tan \phi(r) = u_T/(\Omega r)$.

- (d) Sketch the orientation of the cross section of the blade as r varies from $r = 0$ to $r = \ell$, indicating the x -axis, noting that usually $\Omega\ell > U$. [6]

The lift, L , acts perpendicular to $\mathbf{u}^{\text{eff}}(\mathbf{r})$ on each cross section with a magnitude

$$L = \tilde{c}_L(\alpha)\rho_0|\mathbf{u}^{\text{eff}}(\mathbf{r})|^2 K(r) ,$$

where $K(r)$ is the length of the cross-section of the blade in its longest dimension and $\tilde{c}_L(\alpha)$ is a constant. Note L is a force per unit length of blade.

- (e) Determine the component of the force from the lift parallel to the x -axis. [4]

- (f) Hence show that the length of the cross section, $K(r)$, is [5]

$$K(r) = \frac{2\pi(1 - f)U}{\tilde{c}_L\Omega N} \sin \phi(r) .$$

ANY CALCULATOR

7. (a) Sketch a p-n junction along the direction, x , normal to the junction, with energy perpendicular to x . The junction is in equilibrium at $T = 0$ initially. Include:
- i. The energies of the bottom of the conduction band, $\varepsilon_c(x)$, and the top of the valence band, $\varepsilon_v(x)$, both as a function of x . [2]
 - ii. The energies of the donors and acceptors, indicating which are ionised or neutral. [4]
 - iii. The fermi level. [2]
 - iv. The depletion layer. [2]
- (b) Now turn to a p-n junction under illumination by sunlight under **open-circuit** conditions (i.e. no current is drawn).
- i. Sketch the junction as a function of x (as in last part) now indicating the quasi-fermi levels for the electrons and holes, μ_e and μ_h respectively. [3]
 - ii. Indicate the magnitude of the open circuit voltage, V_{oc} . [2]
 - iii. Explain the physical processes leading to the quasi-fermi levels. [5]
- (c) The diffusion equation is
- $$\frac{\partial n}{\partial t} = D \nabla^2 n$$
- where n is the number density of a particle and D is the diffusion constant. Recombination takes place on a time scale τ . Explain how these quantities and ideas place a limitation on the extent (in x) of the useful region of a p-n junction. [5]
- (d) Discuss how the above ideas determine the overall scale and shape of a photovoltaic device, especially in relation to the absorption length of light in Si being comparable to, or larger than, the length scales mentioned previously in the question. [5]

Formula Sheet

Stefan-Boltzmann law

The flux, F , of energy per unit area per unit time from a body at temperature T with emissivity $0 \leq e \leq 1$ is

$$F = e\sigma T^4,$$

where σ is the Stefan-Boltzmann constant.

Solar flux

At the distance of the Earth from the sun, the radiative flux of energy from the Sun is 1100 Wm^{-2} at sea level.

Bernoulli's formula for steady flow

$$\frac{1}{2}u^2 + \frac{p}{\rho_0} + V = \text{constant}$$

along a streamline, where u is the velocity of the fluid, p is the pressure, ρ_0 is the (fixed) density and V is the potential energy per unit mass per unit volume.

Stokes' theorem

For a vector field $\mathbf{v}(\mathbf{r})$, with an open surface S with boundary C .

$$\int_C d\mathbf{r} \cdot \mathbf{v} = \int_S d\mathbf{S} \cdot \nabla \times \mathbf{v}.$$

Gauss' theorem

For a vector field $\mathbf{v}(\mathbf{r})$ with a closed surface S enclosing a volume V :

$$\int_S \mathbf{v} \cdot d\mathbf{S} = \int_V dV \nabla \cdot \mathbf{v}.$$

Typical wavelength of light in the visible spectrum $\simeq 500 \text{ nm}$

Indirect band gap of Si $\simeq 1.1 \text{ eV}$

Thermal energy room temperature $k_B T \simeq 25 \text{ meV}$ at room temperature $\simeq 300 \text{ K}$.

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.